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RESEARCH ARTICLE

Hybridization between two gartersnake species (*Thamnophis*) of conservation concern: a threat or an important natural interaction?

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Abstract Distinguishing between hybrid zones formed by secondary contact versus parapatric divergence-with-gene-flow is an important challenge for understanding the interplay of geographic isolation and local adaptation in the origin of species. Similarly, distinguishing between natural hybrid zones and those that formed as a consequence of recent human activities has important conservation implications. Recent work has demonstrated the existence of a narrow hybrid zone between the plains gartersnake (*Thamnophis radix*) and Butler's gartersnake (*T. butleri*) in the Great Lakes region of North America, raising questions about the history and conservation value of genetically admixed populations. Both taxa are of conservation concern, and it is not clear whether to regard hybridization as a threat or a natural interaction. Here we use phylogeographic and population genetic methods to assess the timescales of divergence and hybridization, and test for evidence that the hybrid zone is of recent origin. We assayed AFLP markers and ND2 mitochondrial DNA (mtDNA) sequences from *T. radix*, *T. butleri*, and the closely related short-headed gartersnake (*T. brachystoma*) throughout their North American ranges. We find shallow mtDNA divergence overall and high levels of variation within the contact zone. These patterns are inconsistent with recent contact of long-diverged taxa. It is not possible to distinguish true divergence-with-gene-flow from a long-term secondary contact zone, but we infer that the hybrid zone is a long-standing, natural interaction.

Keywords AFLPs · Conservation genetics · Hybrid zone · mtDNA · *Thamnophis*

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Introduction

Hybridization (interbreeding between genetically distinct populations; Harrison 1993) can severely affect the status and recovery of threatened and endangered taxa (Rhymer and Simberloff 1996). Conservationists are often concerned about genetic swamping, or loss of distinctiveness owing to hybridization and gene flow (Rhymer et al. 1994; Wolf et al. 2001; Allendorf and Luikart 2007). This is a particularly important consideration in cases of hybridization between introduced and native species (Allendorf et al. 2001). However, even natural hybridization can indirectly influence threatened and endangered species by affecting the conservation status and legal protection of genetically mixed individuals or populations (O'Brien and Mayr 1991; Allendorf et al. 2001; Schwartz et al. 2004; Haig and Allendorf 2006). Coming to grips with the ethical and legal questions

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raised by natural hybridization is an important challenge for conservation biology.

Few government agencies have clear-cut policies defining the legal status of individuals whose ancestry includes both a protected and unprotected species (Haig and Allendorf 2006). As a result, the consequences of hybridization for conservation are determined on a case-by-case basis. Ideally decisions are informed by scientific research regarding the impact of humans on the hybridization process (e.g. is it a recent consequence of habitat modification; Anderson 1948), the fitness effects of hybridization (Fitzpatrick and Shaffer 2007; Muhlfeld et al. 2009), and impacts of hybrid genotypes on third-party species in native communities (Ayres et al. 2004; Ryan et al. 2009). Here we use geographic analysis of mtDNA and AFLP variation to address questions about the origin of a hybrid zone between two native snakes (both of conservation concern), and the potential value of mixed populations as reservoirs of genetic variation. While mtDNA sequence data provides a historical perspective, its haploid and nonrecombinant mode of transmission make it impossible to address questions about hybridization without additional markers. Given this, we choose to use AFLPs to broadly sample the nuclear genome (e.g., Creer et al. 2004; Savolainen et al. 2006; Fitzpatrick et al. 2008; Nosil et al. 2009). We weigh the evidence for recent secondary contact and admixture vs. a long-standing, natural hybrid zone, and evaluate the genetic variability of the threatened populations in the region of the contact zone. Specifically, we were interested in hybridization between Butler's gartersnake (*Thamnophis butleri*) and the plains gartersnake (*T. radix*).

Morphological and molecular evidence identify *T. radix* as the sister group to *T. butleri* (Rossman et al. 1996; Alfaro and Arnold 2001; de Queiroz et al. 2002) and Rossman et al. (1996) have even suggested that *T. butleri* is a dwarfed (neotenic) derivative of *T. radix*. *Thamnophis butleri* is primarily found in the Midwest Region of the United States east of Lake Michigan with isolated populations in southeast Wisconsin (Fig. 1; Rossman et al. 1996; Harding 1997). Currently, it is listed as Threatened in Wisconsin (WI) (USA) by the WI Department of Natural

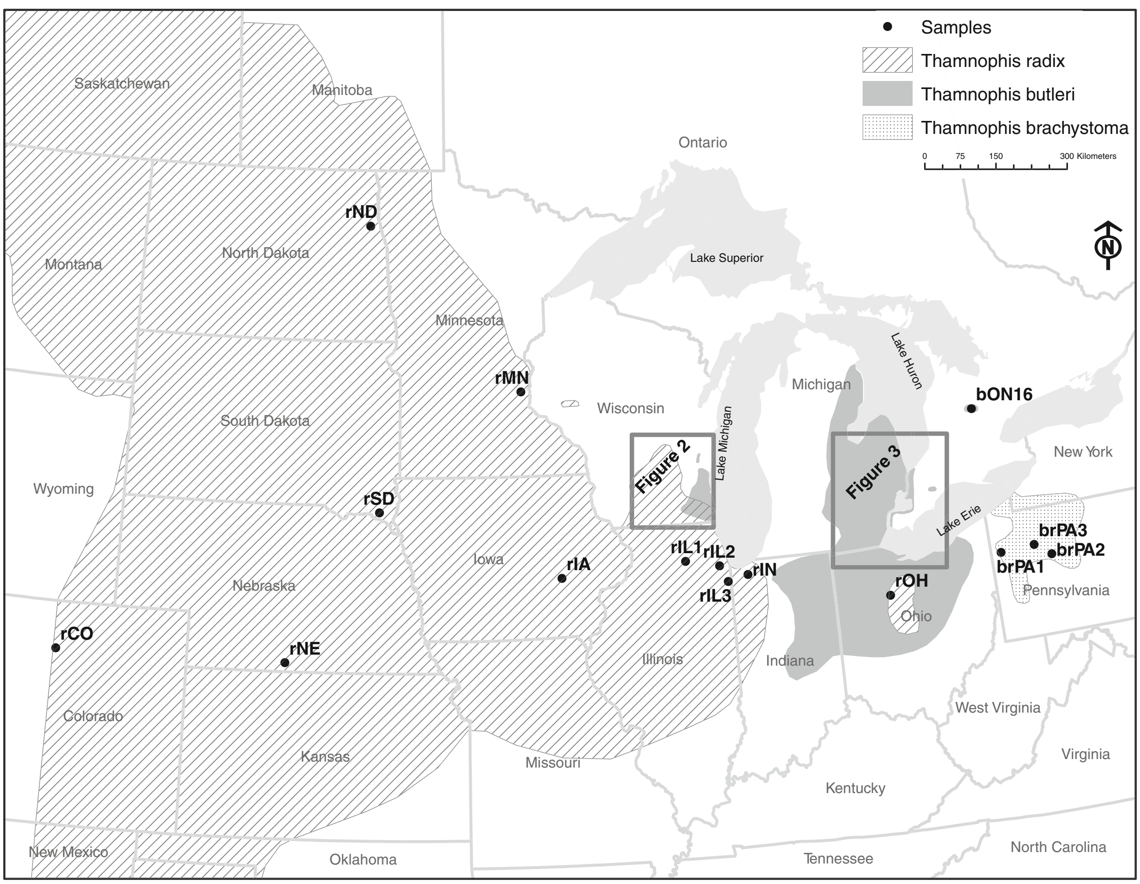

Fig. 1 Sites sampled for Butler's gartersnake (*Thamnophis butleri*), plains gartersnake (*T. radix*), and short-headed gartersnake (*T. brachystoma*) for this study along with the generalized historical range for each species (modified from Rossman et al. 1996). Sample locations for Wisconsin and Ontario/Michigan are shown in Figs. 2 and 3, respectively. Note that *T. butleri* and *T. radix* in Ohio are peripatric, yet there is no evidence that the ranges have ever overlapped. *Thamnophis radix* occupies a very specific habitat type in central Ohio that is not used by *T. butleri*. See text for details. Codes denoting the location of each site are from "Appendix"

Resources (WDNR) (dnr.wi.gov) and Endangered in Indiana (USA) by the Indiana Department of Natural Resources (www.in.gov/dnr/) and Ontario (Canada) by the Ontario Ministry of Natural Resources (www.mnr.gov.on.ca). The decline of *T. butleri* is attributed mainly to habitat destruction, with much of its preferred habitat (wet meadows and prairies) being rapidly developed for commercial and residential purposes. *Thamnophis radix* is widespread in the Great Plains of North America (Fig. 1), but is declining in many locations and is listed as a Species of Special Concern in Wisconsin by the WDNR and Endangered in Ohio (USA) by the Ohio Department of Natural Resources (www.ohiodnr.com) as a consequence of limited habitat availability and habitat destruction (Dalrymple and Reichenbach 1981, 1984; Rossman et al. 1996). In Wisconsin, *T. radix* and *T. butleri* are sympatric and hybridize, but despite coarse-scale range overlap in Ohio, sympatry between the two in that state has never been recorded (Wynn and Moody 2006) most likely due to the lack of suitable prairie habitat for *T. radix* to known populations of *T. butleri*.

Morphological (Casper 2003), behavioral (Ford 1982; Kirby 2005), and molecular data (Fitzpatrick et al. 2008) support the existence of a narrow hybrid zone between Wisconsin populations. Outside of this zone, *T. butleri* and *T. radix* are ecologically and morphologically distinct (Rossman et al. 1996), and even some hybrid populations show evidence of bimodality, indicating that the distinctiveness of the two forms is maintained in the face of gene flow (Fitzpatrick et al. 2008). An important alternative is that secondary contact is very recent, and a process of "species collapse" is just beginning (Taylor et al. 2006; Seehausen et al. 2008). The hybrid zone is coincident with the City of Milwaukee and a vast, growing network of suburbs extending west from Lake Michigan. Therefore, the taxonomically problematic hybrid populations are highly threatened by habitat destruction. A scientifically justified decision regarding their conservation value is an immediate concern.

Methods

Study populations and sample collection

Molecular variation was examined in 549 individual snakes from 74 locations including 316 *T. butleri* from 45 sites, 105 *T. radix* from 17 sites, 123 hybrids from 9 sites within the hybrid zone ("Appendix"; Figs. 1, 2, 3). We also included 5 *T. brachystoma* from 3 sites as outgroups for the mtDNA. In addition to the sequences generated during the course of this study, we also included 1 *T. radix* sequence (GenBank Accession No. AF384853, Alfaro and Arnold 2001), 1 Michigan *T. butleri* sequence (GenBank Accession No. AF420094, de Queiroz et al. 2002) and 1 *T. brachystoma* sequence (GenBank Accession No. AF420091, de Queiroz et al. 2002) from GenBank. While sampling was concentrated in or around the hybrid zone in Wisconsin and Illinois to examine fine scale patterns of variation in this area, additional sites were sampled from across the ranges of both *T. butleri* (i.e. Michigan, Ohio, Ontario) and *T. radix* (i.e. Indiana, Ohio, Iowa, Colorado, Nebraska, Minnesota, North Dakota, South Dakota). Samples were obtained from numerous sources (see Acknowledgments and "Appendix") as frozen muscle tissue or as tail tips or ventral scale clips from live specimens subsequently released at the point of capture.

Mitochondrial DNA amplification

Genomic DNA was obtained with the DNeasy® Tissue Kit (Qiagen). The 985 bases of ND2 that we examined were PCR-amplified using the forward primer L4437b (5′-CAG CTA AAA AAG CTA TCG GGC CCA TAC C-3′; Kumazawa et al. 1996), which lies in the tRNA-Met upstream of ND2 and the reverse primer Sn-ND2r (5′-GGC TTT GAA GGC TMC TAG TTT-3′; R. Lawson, pers. comm.), which lies in the tRNA-Trp downstream of ND2. In each case, polymerase chain reactions (PCR) were conducted in 25-μL volumes with 1.0 μL DNA, 1× ExTaq PCR buffer (PanVera/TaKaRa), 1.5 mM $MgCl_2$, 0.2 mM dNTPs, 0.2 μg/μL bovine serum albumin, 0.1 mM each primer, and 1.25 units of ExTaq polymerase (Panvera/TaKaRa). Amplification conditions involved 30 cycles each consisting of 1 min of denaturing at 94°C, 1 min of primer annealing at 55°C, and 1.5 min of extension at 72°C. PCR products were cleaned prior to sequencing using ExoSAP-IT™ (USB Corporation).

Sequencing reactions were carried out using the internal primers H5382 (5′-GTG TGG GCR ATT CAT GA-3′) and L5238 (5′-ACM TGA CAA AAA ATY GC-3′) (de Queiroz et al. 2002) and Big Dye® Terminator v3.1 Cycle Sequencing kits (Applied Biosystems), and read on an automated sequencer (Applied Biosystems 3100, University of Tennessee Molecular Biology Resource Facility). Sequences were edited using the program Sequencher 3.1.1 (Gene Codes Corporation, Ann Arbor, MI). tRNAs were trimmed from our sequences and alignments were performed initially using Clustal X (Thompson et al. 1997) and subsequently manually refined. Sequences were collapsed into unique haplotypes using Collapse v1.2 prior to analyses.

Mitochondrial DNA analyses

We estimated a mitochondrial gene tree using the ND2 data under the criterion of maximum likelihood (ML) as

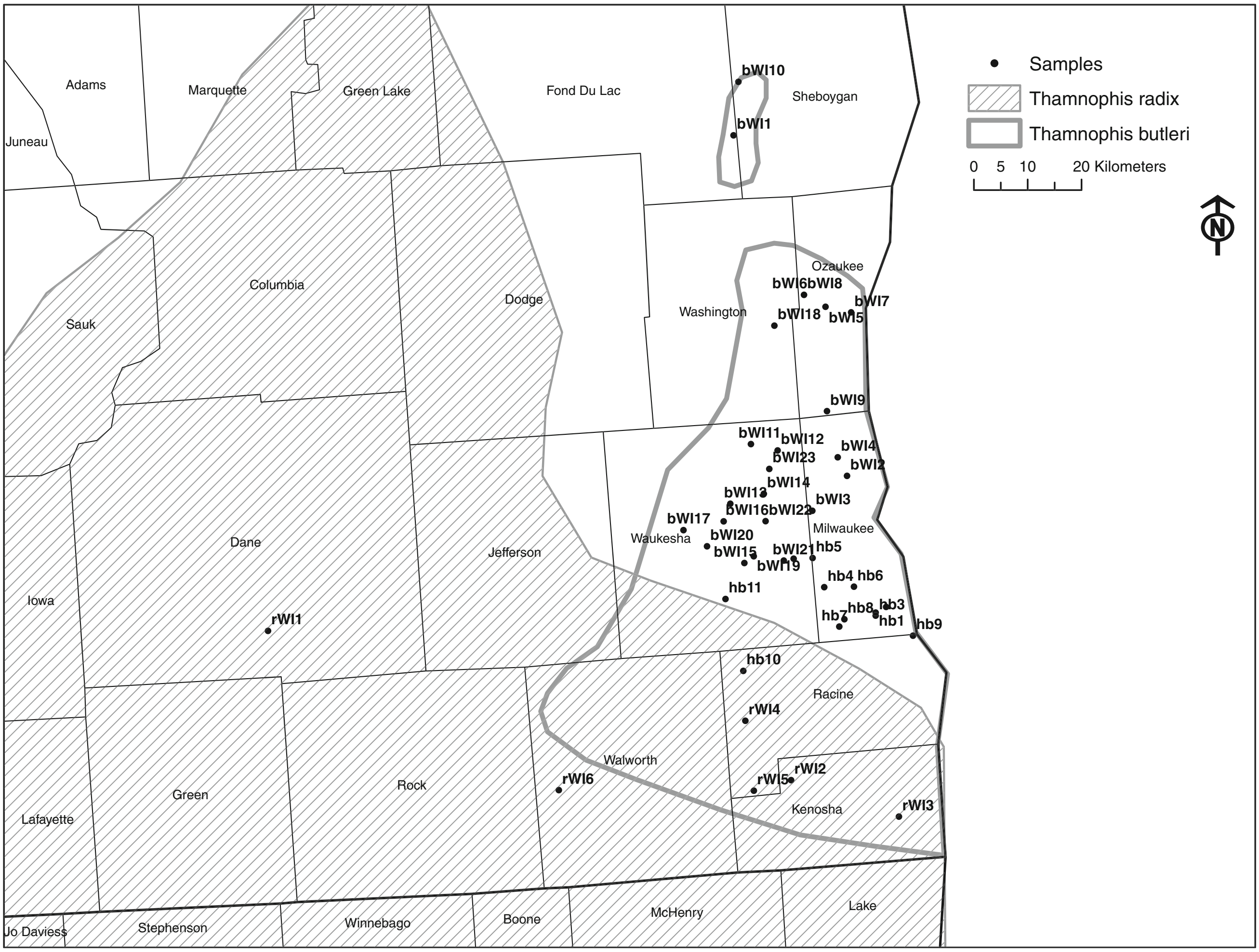

Fig. 2 Wisconsin sites sampled for Butler's gartersnake (*Thamnophis butleri*), plains gartersnake (*T. radix*), and hybrids for this study. *Codes* denoting the location of each site are from "Appendix"

implemented in PAUP* (Swofford 2002) and Bayesian inference of phylogeny (BI) implemented in MrBayes 3.1 (Ronquist and Huelsenbeck 2003). The best fit model of evolution (HKY+G) and estimation of parameters for the dataset under the Akaike information criterion (with correction for small sample size) were determined utilizing Modeltest 3.7 (Posada and Crandall 1998). ML and BI analyses were rooted with published GenBank sequences from *T. elegans*, which represents a clade that is sister to a *T. butleri*, *T. radix*, and *T. brachystoma* clade (Alfaro and Arnold 2001; de Queiroz et al. 2002).

ML analyses were conducted with 1,000 random sequence addition heuristic search replicates with tree-bisection-reconnection (TBR) branch swapping and collapsing all zero-length branches. Bootstrap analysis was employed to assess internal support for the inferred phylogeny using 1,000 bootstrap replicates with simple taxon addition heuristic searches, TBR branch swapping and collapsing all zero-length branches.

All Bayesian analyses were run in duplicate for 5 million generations and were inspected for stationarity (effective mixing and convergence to the posterior distribution) using Tracer v1.4 (Rambaut and Drummond 2007), with the initial 10% of these generations (parameters estimated prior to effective mixing and convergence of the MCMC chain) discarded as burn-in.

Because we sampled many individuals with similar or identical mtDNA haplotypes, we also estimated a haplotype network using TCS 1.13 (Clement et al. 2000) to help visualize the distribution of mtDNA variation. Ambiguous connections (loops or reticulations) in the gene tree were resolved using approaches from coalescent theory (Crandall et al. 1994). In the case of DNA sequence data this resolution generally involves a comparison of the probabilities of whether a haplotype arose via mutation from either a high- or low-frequency haplotype, with frequency evaluated based on both numerical frequency and geographic distribution. That is, abundant and widespread

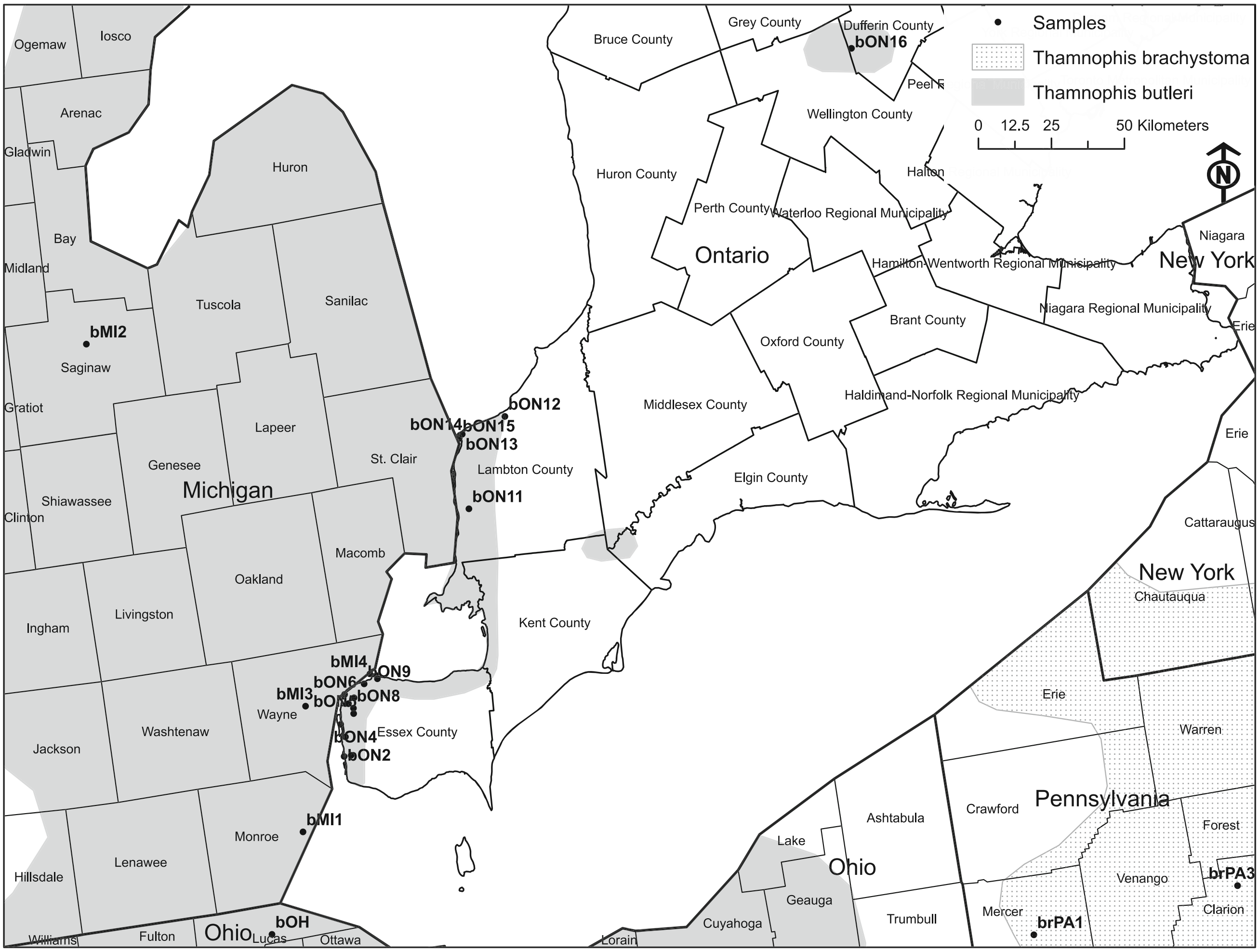

Fig. 3 Ontario, Michigan, Ohio and Pennsylvania sites sampled for Butler's gartersnake (*Thamnophis butleri*), and short-headed gartersnake (*T. brachystoma*) for this study. *Codes* denoting the location of each site are from "Appendix"

haplotypes are more likely ancestors than rare and geographically restricted haplotypes.

mtDNA evidence for population structure was assessed by performing a Spatial Analysis of Molecular Variance (SAMOVA). Φ-statistics were calculated using SAMOVA 1.0 (Dupanloup et al. 2002). SAMOVA consists of identifying groups of populations that are maximally differentiated from each other based on genetic data and spatial relationships. SAMOVA allows for analyses without relying on field identification of individual snakes as either *T. butleri*, *T. radix*, or hybrids based on their morphology. A priori groups, as implemented when using Analysis of Molecular Variance (AMOVA), are not utilized in SAMOVA. However, we also used AMOVAs to examine three specific relationships associated with potential hybridization/recent divergence in Wisconsin: (1) Wisconsin (WI) *T. butleri* vs. WI *T. radix*, (2) WI *T. butleri* vs. hybrids (as identified by morphology; Rossman et al. 1996; Casper 2003), and (3) WI *T. radix* vs. hybrids. *P* values of the Φ-statistics were estimated through 1,000 permutation replicates.

Finally, although there is no explicit phylogeographic test for range expansion into a hybrid zone (see, for example Bloomquist et al. 2010), we used generalized tests of mutation-drift equilibrium to evaluate whether mtDNA variation in the hybrid zone and neighboring sites in Wisconsin gives any indication of recent, non-equilibrium dynamics. We used DnaSP v5 (Rozas and Rozas 1999) to calculate Tajimas *D* and Fu and Li's D^* and $F^*$with associated *P* values. These statistics are significantly less than zero when there are many rare variants, as expected following population growth or a selective sweep, and significantly greater than zero when rare variants are uncommon, as expected from balancing selection or following a persistent bottleneck (Tajima 1989; Fu and Li 1993). Range expansion and admixture would most likely cause negative D, D^* and F^* in populations adjacent to the contact zone, but positive test statistics in recently admixed populations. Structured populations at mutation-migration-drift equilibrium tend to have more rare variants than unstructured populations, hence negative test statistics for all three analyses (Peter et al. 2010).

AFLP analyses

AFLP markers were amplified and scored as described previously (Fitzpatrick et al. 2008). Samples included 14 *T. butleri* from Michigan (MI), 10 from Ohio (OH), 89 from WI north of the hybrid zone (as defined by Casper 2003 and Fitzpatrick et al. 2008), 62 *T. butleri x radix* hybrids from the hybrid zone, 36 *T. radix* from WI (outside the hybrid zone), 13 from OH, and 38 from other states. One or two individuals from each of these groupings were re-run as standards in each 96-well PCR plate to ensure that AFLP fragments were scored consistently for all 262 unique individuals. Financial and logistic constraints limited us to this smaller sample of individuals, but the sample is representative of the geographic ranges of the taxa and the Wisconsin hybrid zone.

To evaluate overall population structure, we performed factorial correspondence analysis (FCA) on fragment presence/absence data (Belkhir et al. 2004) and model-based estimation of admixture proportions using STRUCTURE 2.3.2.1 (Pritchard et al. 2000; Falush et al. 2003). We computed FCA with the MCA function in R 2.11.1 (MASS library version 7.3–6; Venables and Ripley 2002). This analysis uses no a priori information from geography or taxonomy, and simply finds orthogonal multivariate axes of maximum spread among the individual samples. We then overlaid a priori groupings on scatter plots of FCA scores to qualitatively evaluate their correspondence with molecular distance.

In STRUCTURE we ran admixture models with no a priori grouping information. To estimate the number of ancestral lineages we used the ΔK method (Evanno et al. 2005) using 20 replicate runs (10^5 burn-in and 10^5 sampling generations) of each number of lineages (K) from one to ten. We then graphically compared estimates of individual admixture proportions to the a priori groupings.

To assess whether the pattern of AFLP variation is more consistent with recent secondary contact vs. an ancient, stable hybrid zone, we performed three analyses. First, we searched for unique alleles (fragments or null alleles) in each of the a priori sample groupings. Predictions of recent range expansion are that recently colonized regions will have a subset of the alleles present in source populations, and source populations are likely to harbor unique alleles. Given that zero private alleles were found in any a priori grouping, no formal statistical test was performed. See below for a more subtle test of variability in hybrid vs. putative source populations.

Second, we estimated linkage disequilibria (LD) among markers within each a priori grouping. LD is expected to be high in the initial stages of admixture, but LD among neutral markers should decay toward zero in long-standing hybrid zones and be negligible in primary zones established by divergence-with-gene-flow (Long 1991; Barton and Gale 1993; Futuyma 2009). We used r^2 as a simple metric of association between fragment presence/absence at different markers (Laurie et al. 2007), but given the nature of dominant markers, this is not to be taken as an accurate estimator of genotypic disequilibrium comparable to results for co-dominant data. LD decays by ½ each generation in a panmictic population (Lewontin 1974), so absence of elevated LD in mixed populations would rule out only very recent admixture (<10 generations).

Finally, we performed a custom Monte Carlo test of the null hypothesis that the level of variability within the Wisconsin hybrid zone is equivalent to a random admixture of *T. butleri* and *T. radix* from outside of Wisconsin. If the hybrid zone were a result of recent range expansion, we would expect lower variability than the null hypothesis. For example, there are many markers for which one allele is rare (<5%) in both parental gene pools. Those alleles should often be missing from the hybrid zone if it was recently formed by a small sample of colonists from each taxon. On the other hand, if variability within the hybrid zone is greater than the null, then the contact zone might reside in a reservoir of genetic variation that is not explained by admixture alone. The null hypothesis was simulated by forming a random sample of admixed genotypes to match the sample size and distribution of admixture proportions in the observed Wisconsin sample. We estimated allele frequencies using Zhivotovsky's (1999) Bayesian estimator for dominant marker data, assuming Hardy–Weinberg proportions. Then, for a Wisconsin snake with estimated admixture proportion q (fraction of *T. butleri* ancestry), we randomly drew alleles for each marker from a binomial distribution with probability equal to the weighted average of the parental allele frequencies ($qp_{butleri} + [1 - q]\ p_{radix}$). For each random sample of multilocus genotypes we estimated average gene diversity and compared it to the estimate from the observed data. We rejected the null hypothesis if the observed gene diversity was greater or less than the 95% central range of the simulated gene diversities. AFLP data structure and R code for estimation and simulation have been deposited in Dryad (http://dx.doi.org/10.5061/dryad.bk58j54t).

Results

Sequence variation

We obtained complete ND2 sequences for 317 *T. butleri*, 106 *T. radix*, 123 hybrids and 6 *T. brachystoma*. Thirty-six unique haplotypes (GenBank Accession No. HM630317-HM630351 generated in this study and AF384853 from Alfaro and Arnold 2001) were detected from these 552 sequences. Of the 985 bp used in analyses 39 were variable and 16 were phylogenetically informative. No stop codons or indels were detected in any of our sequences, as expected for functional copies of the ND2 gene. Using the program DnaSP v5 (Rozas and Rozas 1999), we found that 14 of the 39 variable sites represented nonsynonymous substitutions, which may be under selection (Zink 2005) and thus may confound our genealogical and phylogenetic analyses. In fact, when examining the overall dataset, six of these were phylogenetically informative sites (38% of phylogenetically informative sites). However, of those six only one was distributed in populations that were geographically isolated from each other.

"Neutrality" tests did not support rejection of mutation-drift equilibrium for any of the a priori groups: Wisconsin *T. radix* (Tajima's D $= -0.71$, Fu and Li's D* test statistic $= -0.69$, Fu and Li's F* test statistic $= -0.78$, $P > 0.10$); Wisconsin *T. butleri*(Tajima's D $= -0.84$, Fu and Li's D* test statistic $= -1.00$, Fu and Li's F* test statistic $= -1.10$, $P > 0.10$); Wisconsin hybrid (Tajima's D $= -0.56$, Fu and Li's D* test statistic $= -0.95$, Fu and Li's F* test statistic $= -0.97$, $P > 0.10$). Most important there was not even a trend toward positive test statistics in the hybrid zone.

Gene tree analyses

To reduce the computational time ML and BI analyses were performed on the 36 unique haplotypes rather than on all 552 sequences (Fig. 4). The relatively short branch lengths exhibited by this tree indicate a very small number of substitutions/site with very little divergence between most haplotypes. Our haplotypes form two distinct clades with one representing *T. butleri* from Ontario, Michigan, and Ohio (haplotypes 30–33) and the other representing all *T. radix*, all *T. brachystoma* and *T. butleri* from every collection location except Michigan and Ontario. Although the fine scale clustering of haplotypes is not strongly supported by high bootstrap values or posterior probabilities, the overall theme that *T. butleri*, *T. radix*, and *T. brachystoma* are not very divergent from each other is strongly supported, as earlier workers have similarly concluded (de Queiroz et al. 2002). Given the extensive haplotype sharing and low level of sequence divergence overall, it would not be appropriate to estimate a timescale of speciation. However, based on the typically high rate of mtDNA sequence evolution in vertebrates (e.g., Crandall et al. 1994; Austin et al. 2002; Allendorf and Luikart 2007; Placyk et al. 2007; Futuyma 2009), we can say that divergence of mtDNA lineages has been very recent.

Haplotype network

Our haplotype network resulted in one independent network including all 36 ND2 haplotypes for *T. radix*,

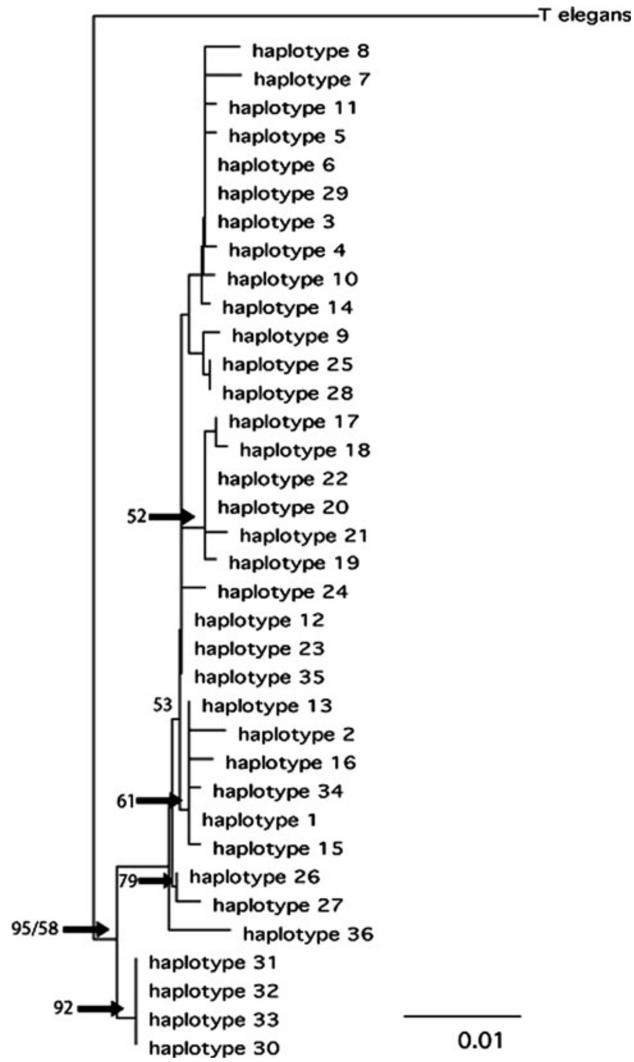

Fig. 4 Maximum likelihood tree recovered from all 36 unique Butler's gartersnake (*Thamnophis butleri*), plains gartersnake (*T. radix*), short-headed gartersnake (*T. brachystoma*), and hybrid ND2 haplotypes. Sites and species represented by each haplotype are listed in "Appendix" and can be visualized in Fig. 5. *Numbers* above branches are bootstrap values (1000 bootstrap replicates with simple taxon addition heuristic searches, TBR branch swapping and collapsing all zero-length branches)/posterior probabilities. Branch lengths are proportional to the expected amounts of character change under the HKY+G model, with the scale as indicated

T. butleri, hybrids, and *T. brachystoma* (Fig. 5; see "Appendix" for sites represented by each haplotype). From this genealogy several important characteristics should be noted. First, 7 of 36 haplotypes were shared between individuals morphologically and geographically diagnosed as *T. butleri* and *T. radix* (haplotypes 1, 3, 4, 20, 22, 23, and 26). Second, 50% of hybrids shared haplotypes with *T. radix* and/or *T. butleri* (haplotypes 1, 3, 4, 16, 19, 20, 23, 24, and 26), with the other 50% exhibiting haplotypes unique to the hybrid zone (haplotypes 9, 10, 11, 12, 13, 14, 17, 18, and 29). Third, all *T. butleri* from Michigan and

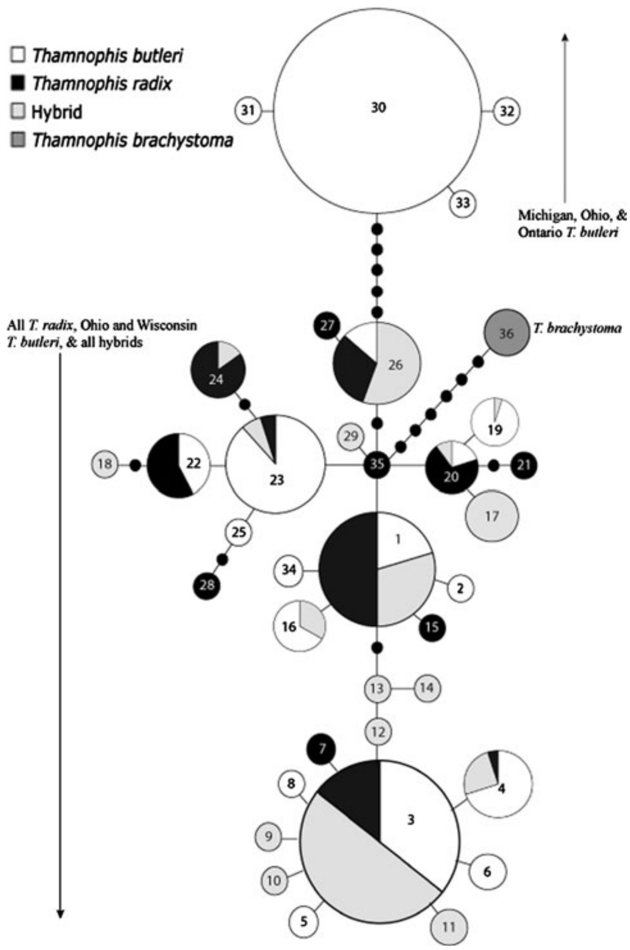

Fig. 5 ND2 haplotype network (Templeton et al. 1992) showing the genealogical relationship of the 36 haplotypes found for 309 Butler's gartersnakes (*Thamnophis butleri*), 105 plains gartersnakes (*T. radix*), 131 hybrids, and 6 short-headed gartersnakes (*T. brachystoma*). *Numbers* indicate individual haplotypes as in "Appendix". *Solid lines* connect haplotypes with a single step. Missing intermediates are indicated by *closed circles* without *numbers*. The *size* of the *circles* corresponds to relative haplotype frequency

Ontario, and some from Ohio (haplotypes 30, 31, 32, and 33) are divergent from the remainder of the genealogy being 6 mutation steps away from the next closest haplotype. Fourth, *T. brachystoma* does not share its haplotype (haplotype 36) with either *T. radix* or *T. butleri* and is 7 mutational steps from matching a *T. radix*-specific haplotype. Finally, the remaining 11 haplotypes are either found only in *T. butleri* (haplotypes 2, 5, 6, 8, 25, and 34) or only in *T. radix* (haplotypes 7, 15, 2, 28, and 35).

21 of the 25 tip haplotypes are specific to one of the four taxonomic units of interest, 2 of the remaining 4 represent sharing between either *T. radix* or *T. butleri* and the hybrids, and 1 represents sharing between *T. radix*, *T. butleri*, and the hybrids. However, most instances where *T. radix*, *T. butleri*, and the hybrids share haplotypes occur with internal haplotypes, possibly indicating that the three diverged from a common ancestor. At the same time our haplotype network confirms how little range wide variation there is, as, for example, haplotype 15, which is found only in a Colorado population and is only one mutational step away from haplotype 1, which is common in the hybrid zone.

SAMOVAs and AMOVAs

When including all 74 *T. butleri*, *T. radix*, hybrid, and *T. brachystoma* sites in the analysis, the SAMOVA with the largest F_{CT} (0.70049), which is associated with the optimal number of simulated groups (Dupanloup et al. 2002), was one with three groups. 70.05% of variation between the groups was best explained by grouping all Michigan and Ontario *T. butleri* together, all remaining *T. butleri*, *T. radix*, and hybrids in a second group, and all *T. brachystoma* in a third group.

In addition to the above SAMOVA we also ran three AMOVAs to examine differences related to the hybridization/recent divergence question in Wisconsin. Comparing WI *T. butleri* to WI *T. radix*, WI *T. butleri* to hybrids, and WI *T. radix* to hybrids, each AMOVA revealed that very little variation was explained among groups with 1.62, 4.52, and 1.49%, respectively, for the three comparisons ($P > 0.05$ for all three). All three AMOVAs indicated that the greatest amount of the Wisconsin variation was explained by within population differences (72.68, 77.95, and 88.55%, respectively) followed by variation among populations within groups (25.7, 17.53, and 9.96%, respectively; $P < 0.05$ for all three).

AFLP population structure

The FCA separated *T. radix* and *T. butleri* along the first axis, with hybrid zone samples largely intermediate (Fig. 6). The second axis clearly separated the Michigan and Ohio *T. butleri* as a distinct cluster of genotypes. Likewise, admixture analysis in STRUCTURE supported a model with three ancestral groups (Table 1). Because the traditional taxonomy (2 species) suggests $K = 2$ a priori, we explicitly compared the $K = 2$ and $K = 3$ cases. When the analysis was limited to $K = 2$ (the prior expectation for admixture between *T. butleri* and *T. radix*), the estimated groupings corresponded to *T. butleri* and *T. radix*, with samples from the hybrid zone showing a range of admixture proportions from 16 to 93% *T. butleri* ancestry (Fig. 7a). When we set $K = 3$ (the best supported K), the Michigan and Ohio *T. butleri* were clustered into a third group, congruent with the mtDNA results (Fig. 7b). These results support the primary distinction between *T. butleri* and *T. radix* with additional subdivision of *T. butleri* between the west and east sides of Lake Michigan.

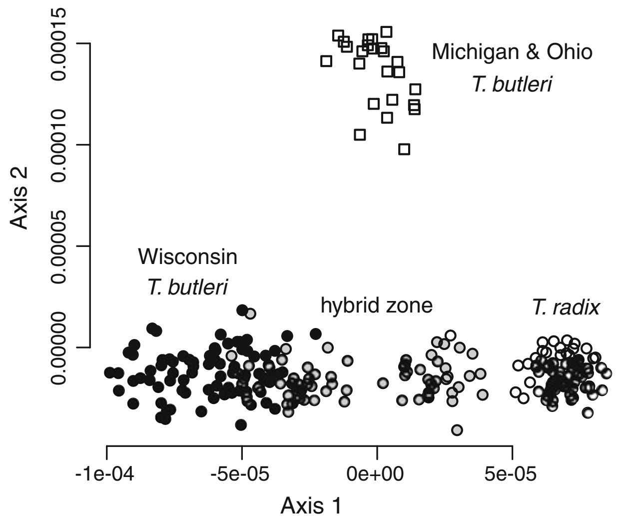

Fig. 6 Ordination of gartersnake genotypes on the first and second axes from a factorial correspondence analysis (FCA) of AFLP band presence/absence

Table 1 Log posterior probabilities, ln Pr(*X*|*K*) (Pritchard et al. 2000), and Δ*K* values (Evanno et al. 2005) for alternative numbers of ancestral genetic clusters (*K*) as estimated from STRUCTURE

K	ln Pr(*X*|*K*)	Δ*K*
1	−71,765.49	NA
2	−66,857.60	60.66
3	−64,066.44	79.44
4	−64,476.35	1.14
5	−64,467.35	0.11
6	−64,504.53	2.65
7	−65,251.78	0.82
8	−65,369.01	0.34
9	−65,207.98	1.12
10	−65,857.69	NA

Unlike the mtDNA, composite analysis of AFLPs shows strong differentiation between *T. butleri* and *T. radix* in Wisconsin, with previously identified hybrid samples generally intermediate (Figs. 6, 7). However, there is no evidence that mtDNA is exceptional. Locus-by-locus analysis of AFLPs revealed a wide range of F_{CT} between Wisconsin *T. butleri* and *T. radix* (Fig. 8), with 86 (26%) markers having lower differentiation than mtDNA (<1.62% of the variation explained by taxon). Estimated allele frequencies in the hybrid zone were intermediate between *T. radix* and *T. butleri* for 314 of 336 variable markers.

The admixture analysis of AFLP markers also suggested a small amount of gene exchange between *T. radix* and *T. butleri* in Ohio (Fig. 7). This might simply reflect uncertainty in the data, but the estimates are consistent with a low level of *T. radix* ancestry in the Ohio *T. butleri* (3–16%).

Tests of recent secondary contact in Wisconsin

There were no AFLP markers unique to WI or hybrid zone populations. Likewise, no markers were entirely absent from WI or hybrid zone populations. Thus, there is no evidence from unique alleles to suggest that WI was recently colonized by either taxon.

Linkage disequilibria (associations of fragment presence/absence between markers) were not elevated in the hybrid zone (Fig. 9), contrary to the expectation of recent secondary contact. Further, average gene diversity in the hybrid zone was significantly greater than the expectation derived from random admixture of *T. butleri* and *T. radix* from outside of Wisconsin ($P < 0.0001$, Table 2), a result opposite of the expected pattern if the hybrid zone was recently established by range expansion of both taxa.

Discussion

We combined mtDNA phylogeography and a survey of AFLP variation across the ranges of *T. butleri* and *T. radix* to test for genetic signatures of recent secondary contact and evaluate the uniqueness and variability of populations in the contact zone. We evaluated four specific predictions about population genetic patterns arising from very recent secondary contact. First, admixture of divergent mtDNA lineages should be recognizable from significant differentiation between parental taxa and positive D, D*, and F* in the hybrid zone. Second, rare private alleles should be more common in source populations than in recently founded hybrid populations. Third, LD should be high in recently founded hybrid populations. Finally, the overall level of variation (controlling for admixture) should be low in the hybrid zone if it was recently established by colonizing populations. None of these predictions were supported by our data. In fact, the level of variability within the hybrid zone was significantly greater than expected from the best fit admixture model. This is further supported by the presence of morphological hybrids collected as early as 1926 (Casper 2003). There is nothing to suggest that habitat alteration in the Milwaukee area has stimulated hybridization only recently. Moreover, high genetic diversity within the hybrid zone and neighboring populations, and the prevalence of unique mtDNA haplotypes in the hybrid zone (50%), indicate that southern Wisconsin is a stronghold of genetic variation and possibly an extant ancestral region from which the distinctive *T. radix, T. butleri*, and *T. brachystoma* lineages arose as they dispersed geographically. Although this conclusion remains

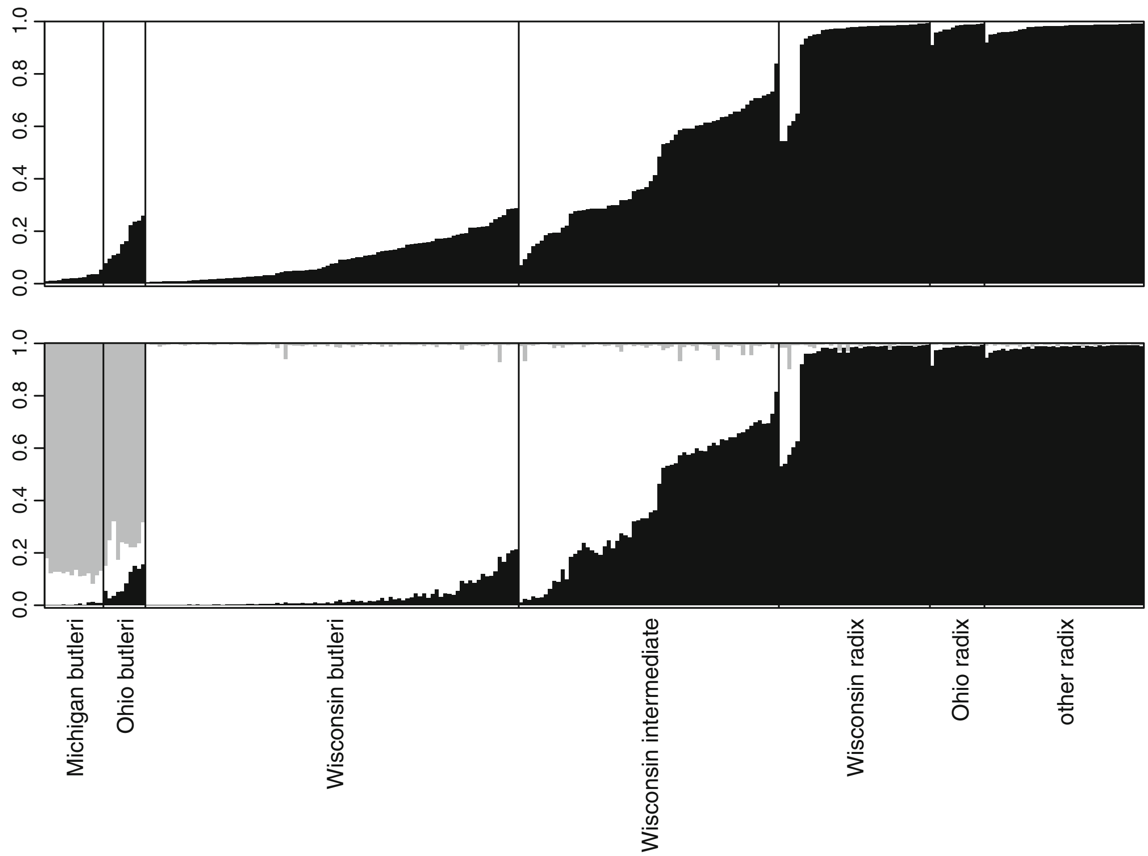

Fig. 7 Admixture proportions estimated for each gartersnake in STRUCTURE with $K = 2$ groups (*top panel*) and $K = 3$ groups (*bottom panel*). A priori groupings were not included in the analysis, but used to order the results. Within each grouping, individuals are sorted by their admixture proportions from the $K = 2$ analysis

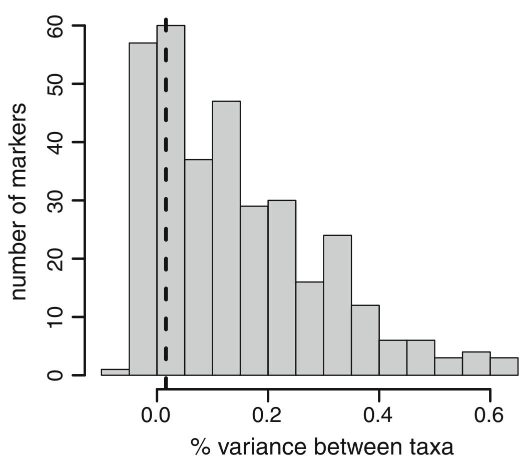

Fig. 8 Distribution of Φ_{CT} (fraction of molecular variance partitioned between taxa) for individual AFLP band presence/absence for WI Butler's gartersnakes (*Thamnophisbutleri*) vs. WI plains gartersnakes (*T. radix*). *Dashed vertical line* illustrates Φ_{CT} for mtDNA sequence data ($\Phi_{CT} = 0.0162$, $P < 0.0001$). The multilocus AMOVA of AFLP distances showed significant differentiation between taxa ($\Phi_{CT} = 0.175$, $P < 0.0001$)

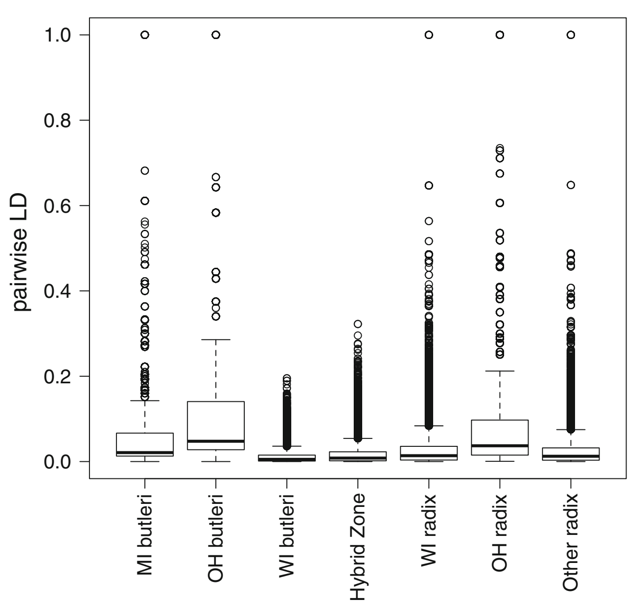

Fig. 9 Distributions of composite genotypic disequilibria (r^2) for pairs of AFLP markers within each a priori group of samples

Table 2 Diploid sample size (n) and average gene diversity (H) for 336 AFLP markers in each a priori grouping of Butler's gartersnake (*Thamnophis butleri*), plains gartersnake (*T. radix*),and hybrids

Group	n	H
MI *T. butleri*	14	0.208
OH *T. butleri*	10	0.264
WI *T. butleri*	89	0.383
Hybrid zone	62	0.359
WI *T. radix*	36	0.284
OH *T. radix*	13	0.303
Other *T. radix*	38	0.267

speculative given the current dataset, future work explicitly testing this hypothesis may be necessary to fully understanding the relationship between these three species of conservation concern, as it suggests a very different attitude toward hybrids and hybrid populations than typically adopted by natural resource managers.

Population structure

Results from mtDNA and AFLP supported the general conclusion that *T. butleri* populations that occur east of Lake Michigan are significantly different from populations that occur west of Lake Michigan, lending further support to previous studies on recolonization of the Great Lakes region following the last glaciation (Austin et al. 2002; Zamudio and Savage 2003; Nice et al. 2005; Placyk et al. 2007). In contrast, *T. radix* showed no geographic structure. Common mtDNA haplotypes occurred throughout its large range, and there was no evidence of subdivision from either AFLP or mtDNA. The pooled samples from 10 states had lower average gene diversity than *T. radix* from Ohio or Wisconsin alone (Table 2).

mtDNA revealed no distinction between *T. butleri* and *T. radix* in Wisconsin. However, this is not unusual; marker-by-marker AMOVAs of the 337 AFLP markers for Wisconsin *T. butleri* vs. *T. radix* revealed little or no differentiation ($P > 0.05$) for 154 markers and lower differentiation than mtDNA (<1.62% of the variance partitioned between taxa) for 86 markers (Fig. 8). In fact, the mtDNA value coincided with modal bin (0.00–0.05) of the AFLP markers. Thus, there is no evidence of particular discordance between mtDNA and nuclear markers; there is simply a great deal of heterogeneity among markers (also see Fitzpatrick et al. 2008; Mims et al. 2010).

Lack of differentiation across the contact zone for a given locus (e.g. mtDNA) could be explained by extensive introgression causing homogenization after secondary contact, lack of differentiation prior to secondary contact, or lack of differentiation owing to ongoing gene flow during primary hybrid zone formation (Endler 1977; Futuyma 2009). The levels of variability and linkage disequilibrium, however, are more germane to the question of whether hybridization between *T. radix* and *T. butleri* is a new interaction precipitated by habitat alteration. In particular, recent admixture is expected to generate genotypic disequilibrium within mixed populations. Disequilibria are not high in the Wisconsin hybrid zone, suggesting that genetic admixture is a long-standing, natural interaction between these native species in southern Wisconsin. In fact, there are many more instances of high pairwise disequilibria outside of Wisconsin. Disequilibrium can arise stochastically owing to drift in small populations and can be perpetuated in growing or expanding populations (Allendorf and Luikart 2007). Thus, if anything, the distributions of genotypic disequilibria among groups (Fig. 9) suggest recent expansion of both *T. butleri* and *T. radix* outside of Wisconsin rather than into Wisconsin. This interpretation is consistent with the overall levels of variability of mtDNA (Fig. 5) and AFLP markers (Table 2).

Hybridization and conservation

Conservation status often depends on taxonomic status, and our results clarify some lingering questions about the relationships between three species of conservation concern. To begin, *T. brachystoma* appears to be the most divergent of the group as evidenced by both our haplotype network and our population-level analyses and should be conserved at the full species status in the two states in which is it listed as a conservation concern. Similarly, it is clear that *T. butleri* from Ohio, Michigan and Ontario constitutes a distinct genetic cluster and should be recognized as an evolutionarily significant unit (ESU). The possibility of gene flow between *T. radix* and *T. butleri* in Ohio merits further study, despite an apparent lack of any modern sympatry (Wynn and Moody 2006). Both the low level of admixture indicated by our results (requiring more than one generation of backcrossing), and the lack of contemporary co-occurrence suggest that hybridization between *T. butleri* and *T. radix* in Ohio must have been an ancient natural interaction.

In Wisconsin, the situation is more complicated. Detailed population-level analyses indicated that *T. butleri* and *T. radix* remain genetically distinct in the face of hybridization (Fitzpatrick et al. 2008). But the legal and biological value of hybrids and admixed populations is difficult to resolve. Based on our results, further consideration should be given to the potential conservation value of the taxonomically problematic hybrid zone populations in the Milwaukee area. Our analyses indicate that gene flow has been occurring for a long time and we infer that the hybrid zone is a natural feature of the group, not an artificial consequence of disturbance. In addition, the hybrid zone appears to harbor more genetic variation than expected under a simple admixture model. This implies that the hybrid zone coincides with a reservoir of relatively ancient genetic variation from which the more widespread and phenotypically distinct populations were derived (e.g. Seehausen et al. 2008). We hypothesize that what is occurring in Wisconsin is speciation in action and that hybrid populations should be conserved at the same level as *T. butleri* populations given their potential importance as sources of genetic variation for ongoing evolution.

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Appendix

See Table 3.

Table 3 Collection data for Butler's gartersnakes (*Thamnophis butleri*), plains gartersnakes (*T. radix*), short-headed gartersnakes (*T. brachystoma*), and hybrids used in mtDNA sequencing and genealogical and phylogenetic analyses

Species	Locality	*n*	Haplotype no.	Lat	Long	Code
Thamnophis brachystoma	Pennsylvania (Mercer Co.)	2	36	41.31	−80.25	brPA1
T. brachystoma	Pennsylvania (Jefferson Co.)	1	36	40.58	−78.52	brPA2
T. brachystoma	Pennsylvania (Venango Co.)	2	36	41.35	−79.34	brPA3
T. butleri	Ohio (Toledo Co.)	21	1, 3, 22, 26, 30, 32*	41.63890000	−83.53650000	bOH
T. butleri	Sterling State Park, Michigan (Monroe Co.)	2	30	41.92145051	−83.34262622	bMI1
T. butleri	Michigan (Saginaw Co.)	1	30	43.40440000	−84.01670000	bMI2
T. butleri	Michigan (Wayne Co.)	7	30, 31*	42.28160000	−83.25730000	bMI3
T. butleri	Belle Isle, Michigan (Wayne Co.)	11	33*	42.34893554	−82.95608587	bMI4
T. butleri	Wisconsin (Fond Du Lac Co.)	2	23	43.65	−88.16	bWI1
T. butleri	Wisconsin (Milwaukee Co.)	19	1, 3, 17, 18*, 20, 23, 26	43.10	−87.96	bWI2
T. butleri	Wisconsin (Milwaukee Co.)	2	19, 23	43.05	−88.05	bWI3
T. butleri	Wisconsin (Milwaukee Co.)	10	3, 4, 23	43.13	−87.97	bWI4
T. butleri	Wisconsin (Ozaukee Co.)	11	3, 8*, 23	43.37	−87.97	bWI5
T. butleri	Wisconsin (Ozaukee Co.)	8	3, 19, 20, 23	43.39	−88.02	bWI6
T. butleri	Wisconsin (Ozaukee Co.)	3	23	43.35	−87.91	bWI7
T. butleri	Wisconsin (Ozaukee Co.)	5	3, 19	43.20	−87.99	bWI8
T. butleri	Wisconsin (Sheboygan Co.)	2	23, 25*	43.73	−88.14	bWI9
T. butleri	Wisconsin (Waukesha Co.)	10	4, 5, 19, 23	43.16	−88.18	bWI10
T. butleri	Wisconsin (Waukesha Co.)	16	3, 4, 5, 19	43.15	−88.12	bWI11
T. butleri	Wisconsin (Waukesha Co.)	19	1, 3, 23	43.07	−88.24	bWI12
T. butleri	Wisconsin (Waukesha Co.)	1	23	43.08	−88.16	bWI13
T. butleri	Wisconsin (Waukesha Co.)	4	2*, 3, 19	42.98	−8.22	bWI14
T. butleri	Wisconsin (Waukesha Co.)	3	6*	43.04	−88.26	bWI15
T. butleri	Wisconsin (Waukesha Co.)	1	1	43.035	−88.36	bWI16
T. butleri	Wisconsin (Waukesha Co.)	9	1, 3, 23	42.98	−88.10	bWI17
T. butleri	Wisconsin (Waukesha Co.)	10	23, 26	42.99	−88.20	bWI18

Table 3 continued

Species	Locality	*n*	Haplotype no.	Lat	Long	Code
T. butleri	Wisconsin (Waukesha Co.)	3	3, 34*	43.01	−88.31	bWI19
T. butleri	Wisconsin (Waukesha Co.)	10	1, 3, 16	42.97	−88.13	bWI20
T. butleri	Wisconsin (Waukesha Co.)	1	3	43.039	−88.16	bWI21
T. butleri	Wisconsin (Waukesha Co.)	3	3, 19	43.12	−88.14	bWI22
T. butleri	Wisconsin (Washington Co.)	11	3, 19, 23	43.34	−88.10	bWI23
T. butleri	Ontario, Canada	1	30	42.12	83.08	bON1
T. butleri	Ontario, Canada	6	30	42.12	83.11	bON2
T. butleri	Ontario, Canada	2	30	42.17	83.10	bON3
T. butleri	Ontario, Canada	15	30	42.21	83.11	bON4
T. butleri	Ontario, Canada	2	30	42.24	83.05	bON5
T. butleri	Ontario, Canada	4	30	42.29	83.07	bON6
T. butleri	Ontario, Canada	1	30	42.33	82.92	bON7
T. butleri	Ontario, Canada	5	30	42.25	83.04	bON8
T. butleri	Ontario, Canada	4	30	42.32	82.98	bON9
T. butleri	Ontario, Canada	8	30	42.28	83.04	bON10
T. butleri	Ontario, Canada	7	30	42.78	82.41	bON11
T. butleri	Ontario, Canada	1	30	43.03	82.20	bON12
T. butleri	Ontario, Canada	3	30	42.99	82.40	bON13
T. butleri	Ontario, Canada	4	30	42.99	82.41	bON14
T. butleri	Ontario, Canada	5	30	42.98	82.41	bON15
T. butleri	Ontario, Canada	22	30	43.93	80.40	bON16
T. butleri	Ontario, Canada	21	30	42.27	83.06	bON17
Hybrid	Wisconsin (Milwaukee Co.)	53	1, 3, 4, 17, 26, 29*	42.88	−87.91	hb1
Hybrid	Wisconsin (Milwaukee Co.)	10	1, 3, 11*, 19, 26	42.93	−88.03	hb2
Hybrid	Wisconsin (Milwaukee Co.)	9	1, 3, 16	42.97	−88.06	hb3
Hybrid	Wisconsin (Milwaukee Co.)	8	1, 3, 4, 23	42.92	−87.96	hb4
Hybrid	Wisconsin (Milwaukee Co.)	10	1, 3, 26	42.86	−88.01	hb5
Hybrid	Wisconsin (Milwaukee Co.)	4	1, 3, 26	42.87	−87.99	hb6
Hybrid	Wisconsin (Racine Co.)	15	1, 3, 4, 12*, 13*, 24, 26	42.84	−87.83	hb7
Hybrid	Wisconsin (Racine Co.)	5	1, 4, 23	42.81	−88.25	hb8
Hybrid	Wisconsin (Waukesha Co.)	9	1, 3, 9*, 10*, 14*, 23, 26	42.92	−88.27	hb9
T. radix	Oregon, Wisconsin (Dane Co.)	3	1	42.92620000	−89.38430000	rWI1
T. radix	Bong Recreation Area, Wisconsin (Kenosha Co.)	25	1, 3, 4, 7, 20, 23, 24, 26, 28*	42.62970000	−88.15210000	rWI2
T. radix	WE Energies Pleasant Prairie Power Plant, Wisconsin (Kenosha Co.)	10	3, 23, 24, 26	42.55829067	−87.90079468	rWI3
T. radix	Honey Creek, Wisconsin (Racine Co.)	5	1, 26	42.72850000	−88.25040000	rWI4
T. radix	Karcher Marsh, Wisconsin (Racine Co.)	1	27*	42.61790000	−88.24370000	rWI5
T. radix	Turtle Creek, Wisconsin (Walworth Co.)	6	3, 26	42.64320000	−88.71190000	rWI6
T. radix	Illinois (DeKalb Co.)	9	1, 20, 21*	41.83450000	−88.71040000	rIL1
T. radix	Illinois (Cook Co.)	5	1, 24, 35*	41.70585394	−87.80448809	rIL2
T. radix	Illinois (Will Co.)	4	1, 3, 24	41.41289867	−87.60847358	rIL3
T. radix	Indiana (Porter Co.)	7	1	41.50560000	−87.06470000	rIN
T. radix	Ohio (Wyandot Co.)	11	22	40.85	−83.34	rOH
T. radix	Iowa (Iowa Co.)	3	1	41.66250000	−92.06650000	rIA
T. radix	Colorado (Boulder Co.)	4	1, 15*	40.15120000	−105.50100000	rCO
T. radix	Nebraska (Harlan Co.)	5	1	40.16660000	−99.45620000	rNE

Table 3 continued

Species	Locality	*n*	Haplotype no.	Lat	Long	Code
T. radix	Minnesota (Ramsey Co.)	1	1	45.01740000	−93.01510000	rMN
T. radix	South Dakota (Clay Co.)	1	1	42.88290000	−97.00680000	rSD
T. radix	North Dakota (Grand Forks Co.)	5	1	48.00380000	−97.35950000	rND

Haplotypes present for each locality are listed with unique haplotypes (i.e. those found only at that locality) indicated by asterisks. Haplotype numbers and codes assigned to each species for each locality correspond with Figs. 1, 2, 3 and 4. Coordinates are generalized for protected Wisconsin and Ontario *T. butleri*, Ohio *T. radix*, and Pennsylvania *T. brachystoma*

References

Alfaro ME, Arnold SJ (2001) Molecular systematics and evolution of *Regina* and the Thamnophiine snakes. Mol Phylogenet Evol 21:408–423

Allendorf FW, Luikart G (2007) Conservation and the genetics of populations. Blackwell, Malden

Allendorf FW, Leary RF, Spruell P, Wenburg JK (2001) The problems with hybrids: setting conservation guidelines. Trends Ecol Evol 16:613–622

Anderson E (1948) Hybridization of the habitat. Evolution 2:1–9

Austin JD, Lougheed SC, Neidrauer L, Chek AA, Boag PT (2002) Cryptic lineages in a small frog: the post-glacial history of the spring peeper, *Pseudacris crucifer* (Anura: Hylidae). Mol Phylogenet Evol 25:316–329

Ayres DR, Smith DL, Zaremba K, Klohr S, Strong DR (2004) Spread of exotic cordgrasses and hybrids (*Spartina* sp.) in the tidal marshes of San Francisco Bay, California, USA. Biol Invasions 6:221–231

Barton NH, Gale KS (1993) Genetic analysis of hybrid zones. In: Harrison RG (ed) Hybrid zones and the evolutionary process. Oxford University Press, New York

Belkhir K, Borsa P, Chikhi L, Raufaste N, Bonhomme F (2004) GENETIX 4.05, logicielsous Windows TM pour la genetique des populations. http://www.genetix.univ-montp2.fr/genetix/genetix.htm

Bloomquist EW, Lemey P, Suchard MA (2010) Three roads diverged? Routes to phylogeographic inference. Trends Ecol Evol 25:626–632

Casper GS (2003) Analysis of amphibian and reptile distributions using presence-only data. PhD dissertation, University of Wisconsin, Madison, Wisconsin

Clement M, Posada D, Crandall KA (2000) TCS: a computer program to estimate gene genealogies. Mol Ecol 9:1657–1659

Crandall KA, Templeton AR, Sing CF (1994) Intraspecific phylogenetics: problems and solutions. In: Scotland RW, Siebert DJ, Williams DM (eds) Models in phylogeny reconstruction. Clarendon Press, Oxford

Creer S, Thorpe RS, Malhotra A, Chou W-H, Stenson AG (2004) The utility of AFLPs for supporting mitochondrial DNA phylogeographical analyses in the Taiwanese bamboo viper, *Trimeresurus stejnegeri*. J Evol Biol 17:100–107

Dalrymple GH, Reichenbach NG (1981) Interactions between the prairie garter snake (*Thamnophis radix*) and the common garter snake (*T. sirtalis*) in Killdeer Plains, Wyandot County, Ohio. Ohio Biol Surv Biol Notes 15:244–250

Dalrymple GH, Reichenbach NG (1984) Management of an endangered species of snake in Ohio, USA. Biol Conserv 30:195–200

de Queiroz A, Lawson R, Lemos-Espinal JA (2002) Phylogenetic relationships of North American garter snakes (*Thamnophis*) based on four mitochondrial genes: how much DNA sequence is enough? Mol Phylogenet Evol 22:315–329

Dupanloup I, Schneider S, Excoffier L (2002) A simulated annealing approach to define the genetic structure of populations. Mol Ecol 11:2571–2581

Endler JA (1977) Geographic variation, speciation, and clines. Princeton University Press, Princeton

Evanno G, Regnaut S, Goudet J (2005) Detecting the number of clusters of individuals using the software structure: a simulation study. Mol Ecol 14:2611–2620

Falush D, Stephens M, Pritchard JK (2003) Inference of population structure using multilocus genotype data: Linked loci and correlated allele frequencies. Genetics 164:1567–1587

Fitzpatrick BM, Shaffer HB (2007) Hybrid vigor between native and introduced salamanders raises new challenges for conservation. Proc Natl Acad Sci USA 104:15793–15798

Fitzpatrick BM, Placyk JS Jr, Niemiller ML, Casper GS, Burghardt GM (2008) Distinctiveness in the face of gene flow: hybridization between specialist and generalist gartersnakes. Mol Ecol 17:4107–4117

Ford NB (1982) Species specificity of sex pheromone trails of sympatric and allopatric garter snakes. Copeia 1982:10–13

Fu YX, Li WH (1993) Statistical tests of neutrality of mutations. Genetics 133:693–709

Futuyma DJ (2009) Evolution, 2nd edn. Sinauer Associates, Inc., Sunderland

Haig SM, Allendorf FW (2006) Hybrid policies under the U.S. Endangered Species Act. In: Scott JM, Gobble DD, Davis F (eds) The endangered species act at thirty: conserving biodiversity in human-dominated landscapes. Island Press, Washington, DC

Harding JH (1997) Amphibians and reptiles of the great lakes region. The University of Michigan Press, Ann Arbor

Harrison RG (1993) Hybrids and hybrid zones: historical perspective. In: Harrison RG (ed) Hybrid zones and the evolutionary process. Oxford University Press, New York

Kirby LE (2005) A comparative study of behavior in neonate gartersnakes, *Thamnophis butleri* and *T. radix* (Colubridae), in an area of potential hybridization. MA thesis, University of Tennessee, Knoxville Tennessee

Kumazawa Y, Ota H, Nishida M, Ozawa T (1996) Gene rearrangements in snake mitochondrial genomes: highly concerted evolution of control-region-like sequences duplicated and inserted into a tRNA gene cluster. Mol Biol Evol 13:1242–1254

Laurie CC, Nickerson DA, Anderson AD, Weir BS, Livingston RJ, Dean MD, Smith KL, Schadt EE, Nachman MW (2007) Linkage disequilibrium in wild mice. PLOS Genetics 3:e144. doi:110.1371/journal.pgen.0030144

Lewontin RC (1974) The genetic basis of evolutionary change. Columbia University Press, New York

Long JC (1991) The genetic structure of admixed populations. Genetics 127:417–428

Mims MC, Hulsey CD, Fitzpatrick BM, Streelman JT (2010) Geography disentangles introgression from ancestral polymorphism in Lake Malawi cichlids. Mol Ecol 19:940–951

Muhlfeld CC, Kalinowski ST, McMahon TE, Taper ML, Painter S, Leary RF, Allendorf FW (2009) Hybridization rapidly reduces fitness of a native trout in the wild. Biol Lett 5:328–331

Nice CC, Anthony N, Gelembiuk G, Raterman D, French-Constant R (2005) The history and geography of diversification within the butterfly genus *Lycaeides* in North America. Mol Ecol 14:1741–1754

Nosil P, Funk DJ, Ortiz-Barrientos D (2009) Divergent selection and heterogeneous genomic divergence. Mol Ecol 18:375–402

O'Brien SJ, Mayr E (1991) Bureaucratic mischief: recognizing endangered species and subspecies. Science 251:1187–1188

Peter BM, Wegmann D, Excoffier L (2010) Distinguishing between population bottleneck and population subdivision by a Bayesian model choice procedure. Mol Ecol 19:4648–4660

Placyk JS Jr, Burghardt GM, Small RL, King RB, Casper GS, Robinson JW (2007) Post- glacial recolonization of the Great Lakes region by the common gartersnake (*Thamnophis sirtalis*) inferred from mtDNA sequences. Mol Phylogenet Evol 43:452–467

Posada D, Crandall KA (1998) MODELTEST: testing the model of DNA substitution. Bioinformatics 14:817–818

Pritchard JK, Stephens M, Donnelly P (2000) Inference of population structure using multilocus genotype data. Genetics 155:945–959

Rambaut A, Drummond AJ (2007) Tracer v1.4. Available from http://beast.bio.ed.ac.uk/Tracer

Rhymer JM, Simberloff DS (1996) Genetic extinction through hybridization and introgression. Annu Rev Ecol Syst 27:83–109

Rhymer JM, Williams MJ, Braun MJ (1994) Mitochondrial analysis of gene flow between New Zealand mallards (*Anas platyrhynchos*) and grey ducks (*A. superciliosa*). Auk 111:970–978

Ronquist F, Huelsenbeck JP (2003) MrBayes 3: Bayesian phylogenetic inference under mixed models. Bioinformatics 19:1572–1574

Rossman DA, Ford NB, Seigel RA (1996) The garter snakes: evolution and ecology. University of Oklahoma Press, Norman

Rozas J, Rozas R (1999) DnaSP version 3: an integrated program for molecular population genetics and molecular evolution analysis. Bioinformatics 15:174–175

Ryan ME, Johnson JR, Fitzpatrick BM (2009) Invasive hybrid tiger salamander genotypes impact native amphibians. Proc Natl Acad Sci USA 106:11166–11171

Savolainen V, Anstett M-C, Lexer C, Hutton I, Clarkson JJ, Norup MV, Powell MP, Springate D, Salamin N, Baker WJ (2006) Sympatric speciation in palms on an oceanic island. Nature 441:210–213

Schwartz MK, Pilgrim KL, McKelvey KS, Lindquist EL, Claar JJ, Loch S, Ruggiero LF (2004) Hybridization between Canada lynx and bobcats: genetic results and management implications. Conserv Genet 5:349–355

Seehausen O, Takimoto G, Roy D, Jokela J (2008) Speciation reversal and biodiversity dynamics with hybridization in changing environments. Mol Ecol 17:30–44

Swofford D (2002) PAUP*: phylogenetic analysis using parsimony (*and other methods), version 4.0b10. Sinauer Associates, Sunderland

Tajima F (1989) Statistical method for testing the neutral mutation hypothesis by DNA polymorphism. Genetics 123:585–595

Taylor EB, Boughman JW, Groenenboom M, Sniatynski M, Schluter D, Gow JL (2006) Speciation in reverse: morphological and genetic evidence of the collapse of a three-spined stickleback (*Gasterosteus aculeatus*) species pair. Mol Ecol 15:343–355

Templeton AR, Crandall KA, Sing CF (1992) A cladistic analysis of phenotypic associations with haplotypes inferred from restriction endonucleases mapping and DNA sequence data. III. Cladogram estimation. Genetics 132:619–633

Thompson JD, Gibson TJ, Plewnial F, Jeanmougin F, Higgins DG (1997) The Clustal X windows interface: flexible strategies for multiple alignment aided by quality analysis tools. Nucleic Acids Res 25:4876–4882

Venables WN, Ripley BD (2002) Modern applied statistics with S-Plus, 4th edn. Springer, New York

Wolf DE, Takebayashi N, Rieseberg LH (2001) Predicting the risk of extinction through hybridization. Conserv Biol 15:1039–1053

Wynn DE, Moody SM (2006) Ohio turtle, lizard, and snake atlas. Ohio Biological Survey, Columbus

Zamudio KR, Savage WK (2003) Historical isolation, range expansion, and secondary contact of two highly divergent mitochondrial lineages in spotted salamanders (*Ambystoma maculatum*). Evolution 57:1631–1652

Zhivotovsky LA (1999) Estimating population structure in diploids with multilocus dominant DNA markers. Mol Ecol 8:907–913

Zink RM (2005) Natural selection on mitochondrial DNA in *Parus* and its relevance for phylogeographic studies. Proc R Soc Biol 272:71–78